

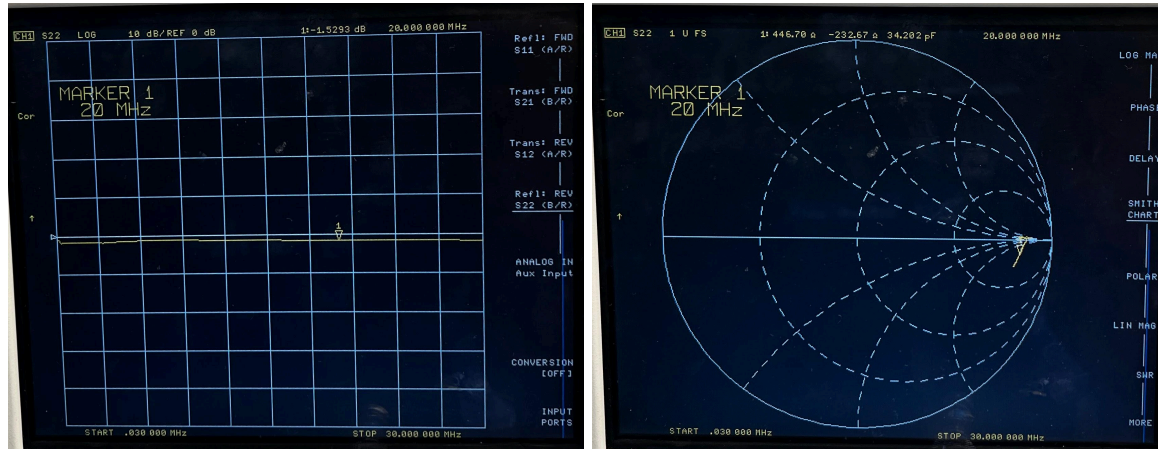
EE 308 Lab 1

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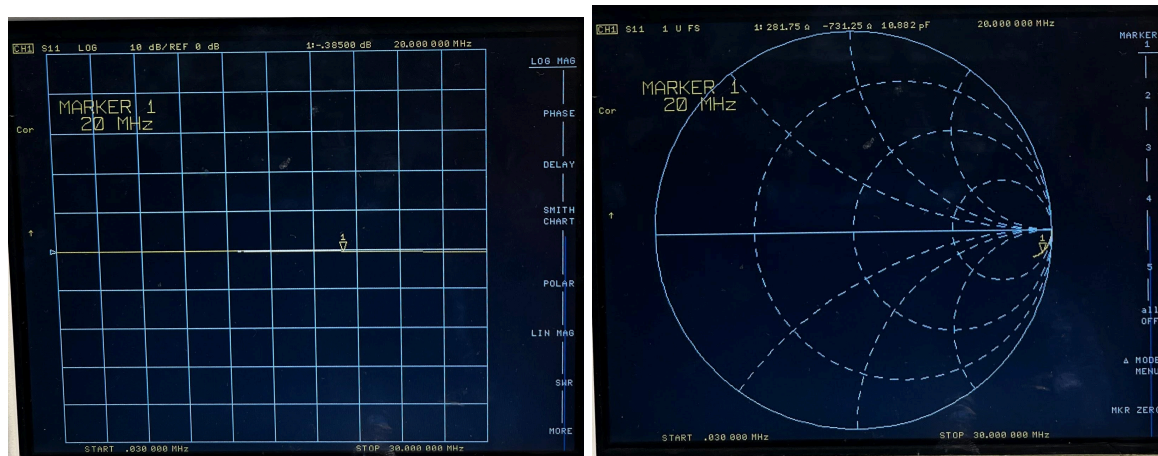
Output impedance

446.7-232.67j across the frequency range of 30kHz to 20MHz



Input Impedance

281.75-731.25j across the frequency range of 30kHz to 20MHz.

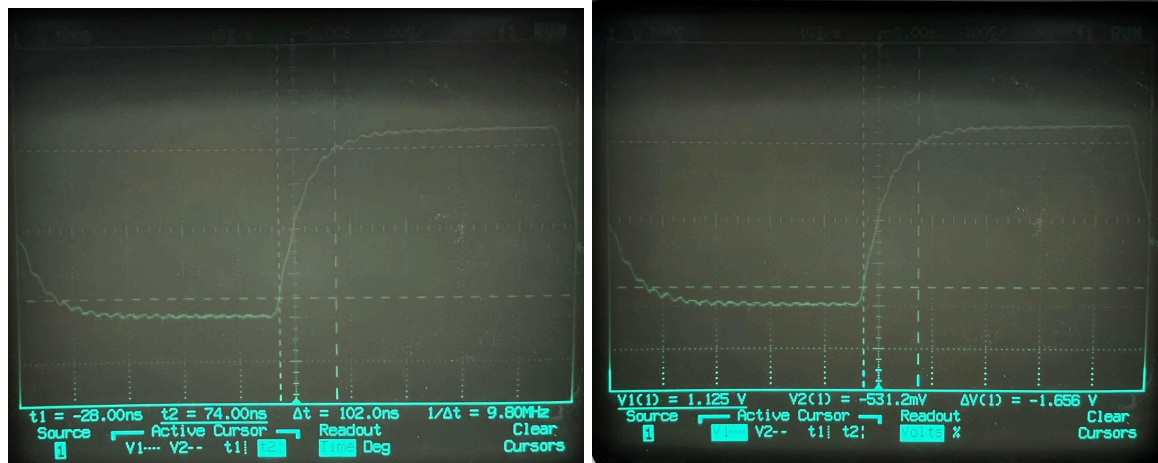


Small signal bandwidth

We measured our bandwidth to be 6MHz while maintaining 10 gain or higher. From the highest frequency we were able to measure was 15MHz which had a gain of 5.

Slew rate

The photos below show the cursor measurements from 10% to 90% of a square wave output from our amplifier. The photos are the same but the first shows the rise time and the second shows the change in voltage. From these values we can compute the slew rate.

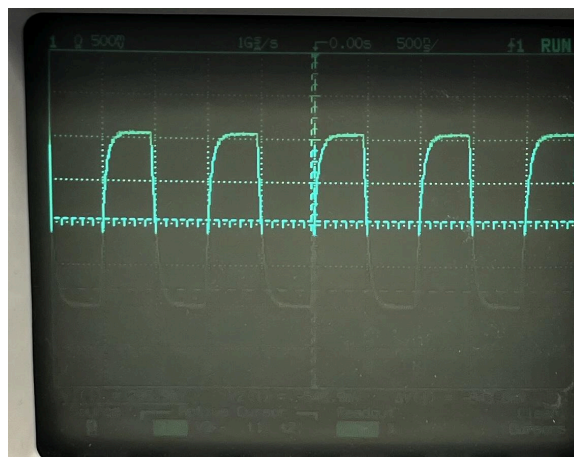


From these images we have $\Delta t = 102\text{ns}$ and $\Delta V = 1.656\text{V}$. Then we can compute our slew rate.

$$\text{slew rate} = \frac{\Delta V}{\Delta t} = \frac{1.656\text{V}}{102\text{ns}} \approx 16\text{kV}/\mu\text{s}$$

Overshoot

We do not experience any overshoot so we have 0% overshoot. The following is an image of the output of our amplifier when inputting a 100mV square wave.



Total Harmonic Distortion

Our measured total harmonic distortion is 1.78%. The following is a photo of the spectrum analyzer when fed a 2MHz sine wave output from our amplifier.



The following is a table of the measured harmonic peaks.

	dBm	Peak Voltage (V)
V1	6.7	0.684
V2	-29.6	0.0105
V3	-34.1	0.00624
V4	-56.8	0.000457
V5	-65	0.000178

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + V_5^2 + \dots}}{V_1} \times 100\%$$
$$THD = \frac{\sqrt{(0.0105)^2 + (0.00624)^2 + (0.000457)^2 + (0.000178)^2}}{0.684} \times 100\% = 1.78\%$$

Schematic

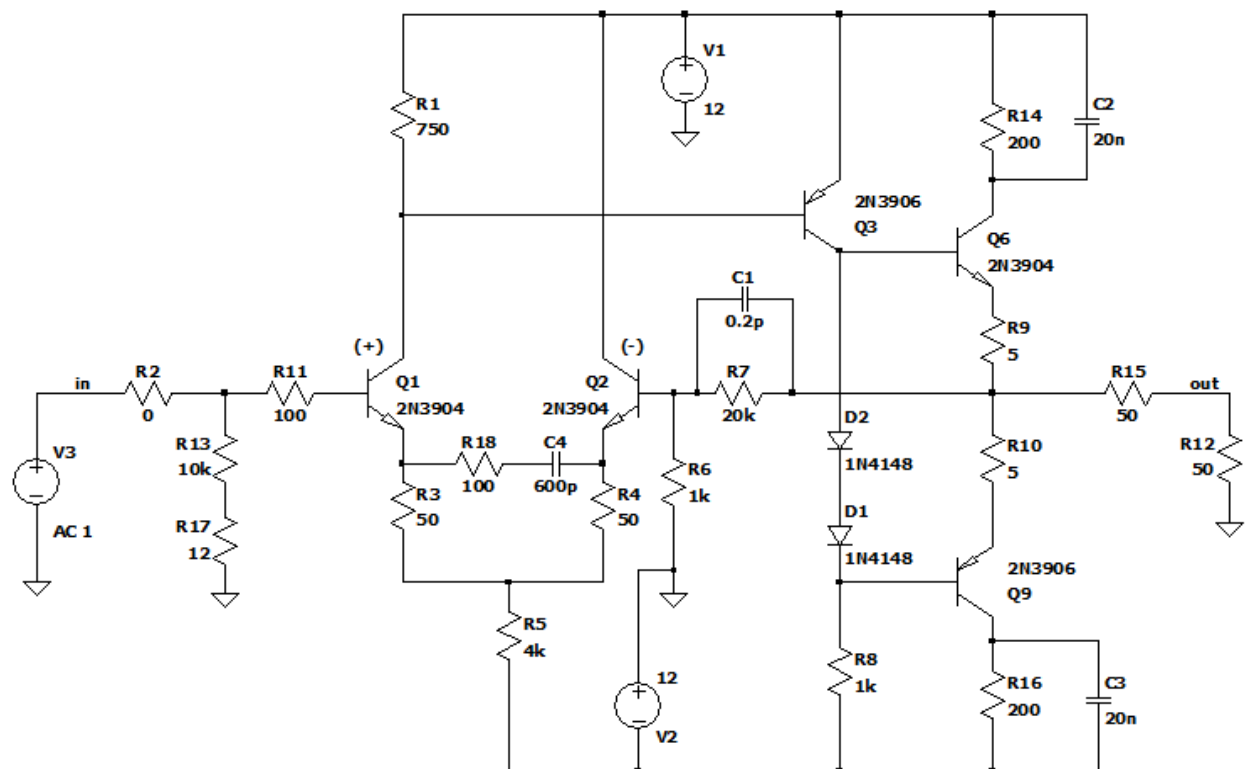
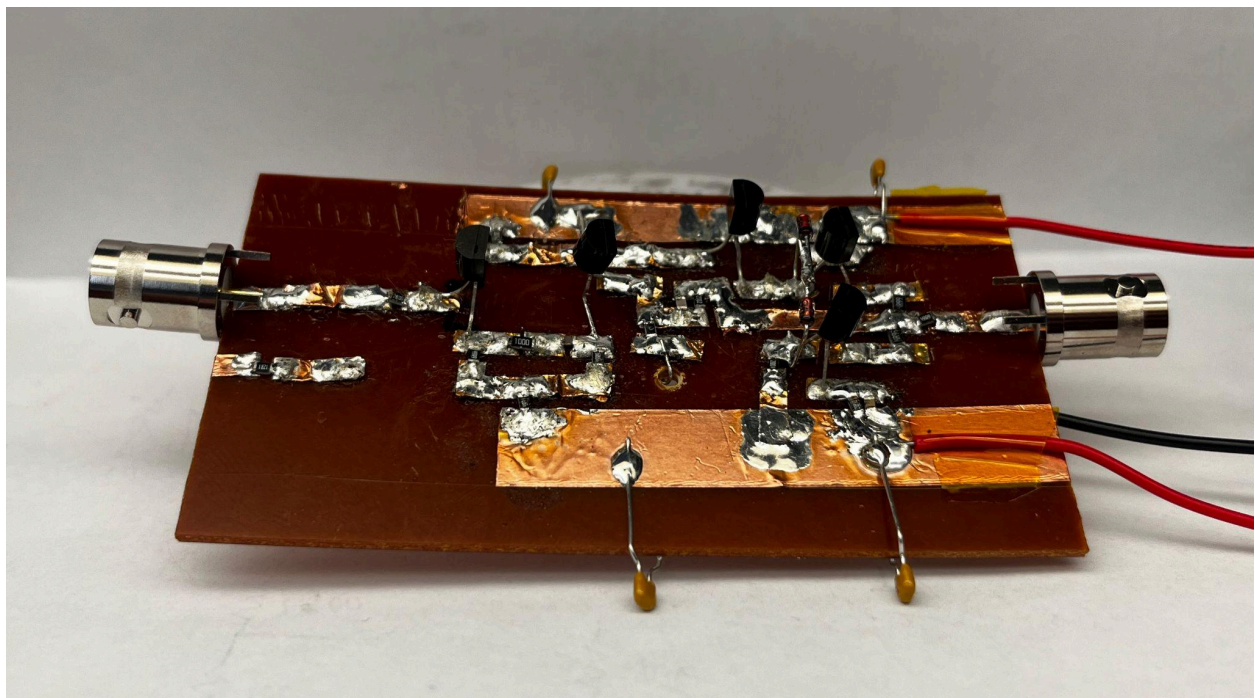


Photo of Board



Discussion of your chosen architecture, design approach, and potential discrepancies between your simulations, calculations, and measurements. Include key calculations you made.

Technical Discussion

Summary:

Our design is based on the BK3011. It starts with a differential pair, followed by an amplifier, and ends with an output driver (push-pull) stage. We wrap this amplifier with feedback to help boost the gain.

Differential Pair Stage:

The differential pair was designed to have a bias current of 3mA. Split between the two transistors this makes the collector current of each 2N3904 roughly 1.5mA. Because the stage that follows is a common emitter amplifier using a 2N3906, we want the output voltage to be close to the rail so the transistor will easily turn on. For this reason we chose the output to swing around 10.5V. This is a 1.5V drop across the emitter resistor of the diff pair. Thus the resistor has the following value.

$$R = 1.5V / 1.5mA = 1k\Omega$$

The input to the differential pair is going to be a sine wave, triangle wave, or square wave centered around zero. Thus we can roughly approximate the emitter voltage to be -0.7V. Then for 3mA current we can compute the tail resistor.

$$R = (12V + 0.7V) / 3mA \approx 4.2k\Omega$$

We then added 50Ω as degeneration to improve stability of the diff pair transistors. A rule of thumb from the Art of Electronics is for the emitter resistor to be less than 100Ω and we found that 50Ω from tuning in simulation. Finally for the diff pair we also added a speed boost using a capacitor in series with a resistor counting the emitters which was tuned in simulation.

Gain Stage and Output Driver:

The gain stage is a common emitter amplifier. We included two diodes in series to help bias the output driver stage BJT's on. This helps with cross over distortion. In terms of the output driver itself, the emitter and collector resistor were placed to prevent thermal runaway at the output. The emitter resistor is 50hm, as larger values of degeneration greatly reduce the gain. In order to add short protection, we used another larger resistor at the collector that if sized properly creates a negative feedback [loop](#) as stated in the BJT Primer document. Worst case scenario, we did not want our BJT's to experience more that 100mA of current, so we calculated the R_C with equation:

$$R_E + R_C = \frac{V_{CC}}{2I_C}$$

At minimum, we needed an R_C of 60 ohms, but we oversized to 200 Ohms with parallel caps. This may come at the cost of of slew rate, but since our slew rate was above spec, we maintained the higher R for increased protection.